8D report

Customer plant:	Nidec Poland	**Supplier:**	AVO Carbon Germany	**Date of claim:**	02.07.2025
Customer contact:	Dominik Dudek	**AVO contact:**	Gunther Luschtinetz		
Customer reference:	503-660-99-99 - 2025/7/2/0360	**AVO reference:**	250703		

QUICK RESPONSE (D1-D3)

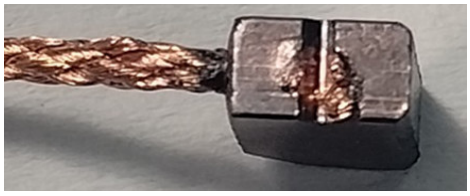
This document must be issued within 24 hours after claim notification

D 1. Description of the problem

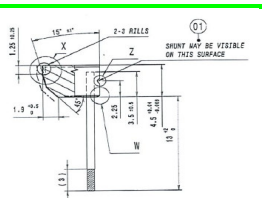
1.a What is the problem? Use 5W+2H to describe what is the current situation.

Problem description	Not proper sintering process, wrong wire position into brush	What happened	Not proper sintering process, wrong wire position into brush
		Why is it a problem	wire come off, brush is craking
Incident type	Customer claim		
Part number AVO	1027700	When did it happen	02.07.2025
Part number Customer	503-660-99-99	Who detected it	Operator
Batch number	282981	Where was it detected	Production
Recurrence	Non-recurrent	How was it detected	Visually
Defect S/R	No	How many bad parts	12

nOK part picture/ sketch



OK part picture/ sketch



1.b Problem solving team

Name	Position	Location	E-mail
Gunther Luschtinetz	Quality Management	AVO Germany	gunther.luschtinetz@avocarbon.com
Daniel Beil	Quality Management	AVO Germany	daniel.beil@avocarbon.com
Mustafa Yorulmaz	Production	AVO Germany	
Martin Röhr	Production	AVO Germany	
Khaled Amri	Logistic	AVO Germany	

1.c Tools to use

Yes	Ishikawa	Yes	5why's		LLC		Audit		Other

1.d First analyse of problem / potential root causes

Potential root cause(s): Brass bushings worn

If you use our expressnumber, please use only economic express.

Samples of nok parts required in plant for analyse: Yes

Express shipping number account: FedEx 186141300

D 2. Risk on similar products and processes

Can this defect happen on similar parts?

		Comments (compulsory, if yes)
Other family product	No	
Other line	No	
Other tools/work station	No	
Other	No	

D 3. Containment

Sorting criterias (add additional sheets if needed)

Location in the supply chain	N° of involved parts	N° parts sorted	N° nok parts	Ppm	Date	Responsible
Parts at customer	5.000	---	12	---	04.07.2025	Dominik Dudek
Parts in transport	---	---	---	---	---	---
Parts at AVO Germany stock	---	---	---	---	---	Gunther Luschtinetz
Parts at AVO production	---	---	---	---	---	---

First contained shipment information

Immediate actions to prevent production and delivery of nok parts ?

Containment actions	Responsible	Date planned	Date closed
Register the complaint + Quality alert	Gunther Luschtinetz	04.07.2025	04.07.2025
Check the production-, quality- and maintenance data	Gunther Luschtinetz	04.07.2025	04.07.2025
Information of all department leader during shop floor meeting QRQC	Joachim Gehl	04.07.2025	04.07.2025
Claimed box (5000pcs) send back to AVO for analysis	Gunther Luschtinetz	04.07.2025	04.07.2025
Agree a sorting of the 45.000 parts	D. Dudek	04.07.2025	04.07.2025

8D report

PLAN - DO (D4-D6)

D 4. Root cause of occurrence

Work out the root causes using Ishikawa and 5 why's

Root cause of occurrence :

The guide of the scissors, which consists of brass bushings, was worn. As a result, the scissors moved away when cutting the wire and were not always in the same position.
This led to a different wire pressing direction in the further course of the process.
If the wire is pressed too far toward the groove, damage can occur during the subsequent grinding process because the grinding tool becomes coated because of the copper.

How was the defect reproduced ?

With the worn brass bushings.

D 5. Root cause of non detection

Work out the root causes using Ishikawa and 5 why's

Root cause of non detection :

Risk to the grinding tool and thus to the brush was not sufficiently considered.

D 6. Corrective actions and non detection

Corrective actions	Responsible	Date planned	Date closed
Replace the worn brass bushings	G. Dohr	09.07.2025	16.07.2025
Implement the bushings in the checklist	G. Luschtinetz	16.07.2025	16.07.2025
Update P-FMEA	Team	19.07.2025	20.07.2025

CHECK - ACT (D7-D8)

D 7. Effectiveness of action plan

Actions to check effectiveness (to plan with part D6)	Responsible	Date planned	Date closed
Check the position of the wire in the brush	Q-Team	16.07.2025	19.07.2025

D 8. Lessons learned

8.1 Do following documents require an update ?

	Responsible	Date planned	Date closed
P-FMEA	Team	19.07.2025	20.07.2025
Control Plan			
Procedure			
Instructions			
Other			
Other			

8.2 Generalization of countermeasures

Checklist is valid for all brushes.

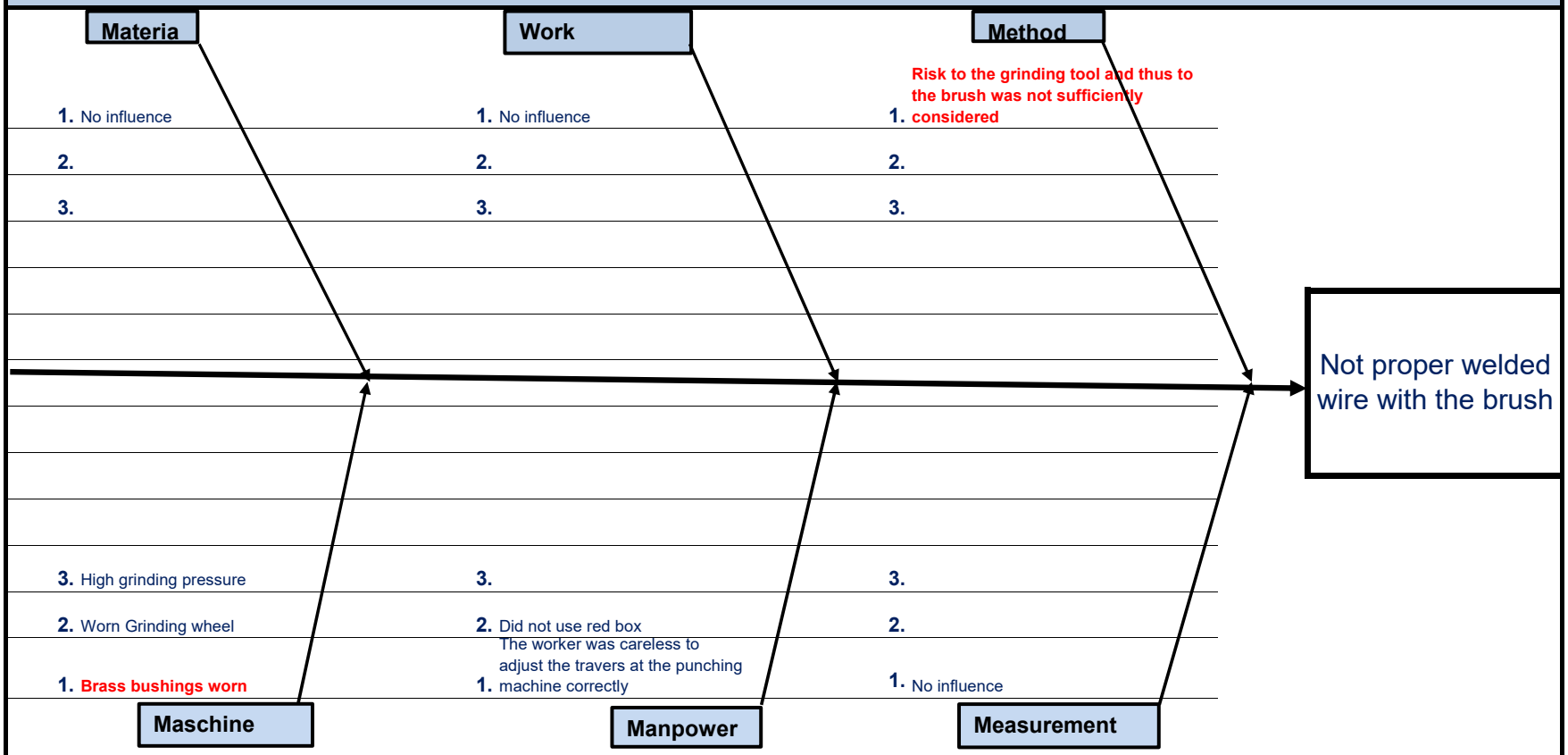
Generally, if one of the tools listed above is used. This is carefully checked and updated.

8.3 Internal audit (if required)

Date 20.07.25 Results ok. No deviation found

Comments

Ishikawa


Materia
1. No influence
2.
3.
3. High grinding pressure
2. Worn Grinding wheel
1. Brass bushings worn
Maschine
Work
1. No influence
2.
3.
3.
2. Did not use red box
1. The worker was careless to adjust the travers at the punching machine correctly
Manpower
Method
1. Risk to the grinding tool and thus to the brush was not sufficiently considered
2.
3.
3.
2.
1. No influence
Measurement
Not proper welded wire with the brush

5 Why's

Problem description : Not proper welded wire with the brush

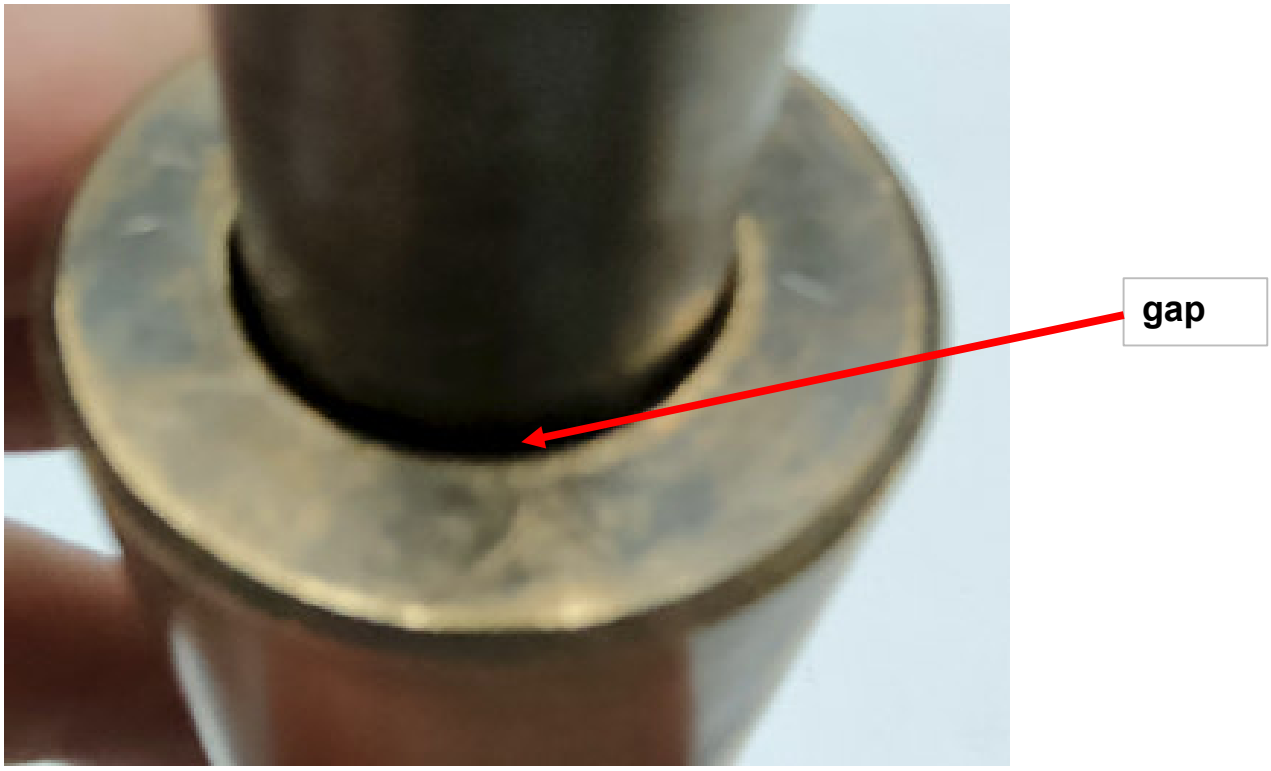
OCCURENCE

Problem	N°	1st WHY?	1st ANSWER	2nd WHY?	2nd ANSWER	3rd WHY?	3rd ANSWER	4th WHY?	4th ANSWER	5th WHY?	5th ANSWER
	1	Why is the carbon brush damaged?	Because the grinding pressure has increased.	Why has the grinding pressure increased?	Because the tool was coated with copper.	Whyl was the tool coated with copper?	Because the wire was pressed too much towards the groove.	Why was the wire pressed too much towards the groove?	Because the scissors had different positions when cutting.	Why had the scissors different positions when cutting.	Because the guide bushings on the scissors were worn out.

NON-DETECTION

Problem	N°	1st WHY?	1st ANSWER	2nd WHY?	2nd ANSWER	3rd WHY?	3rd ANSWER	4th WHY?	4th ANSWER	5th WHY ?	5th ANSWER
	1	Why was the defect not detected?	Because the defect was not recognized as such.	Why was the defect not recognized as such?	Because wire may be visible and this was transferred to the pressing process.	Why was this transferred to the pressing process?	Because the risk to the grinding tool and thus to the brush was not sufficiently considered.				

D 4. Root cause of occurrence



D 6. Corrective actions and non detection

Implement the bushings in the checklist

Produktionsstart

Produktionsende

Typwechsel (Einrichter) & Loswechsel (Bediener)		(NA = Nicht Anwendbar)				
		Start		Ende		NA
Checkliste	Vorgehen	Ja	Nein	Ja	Nein	
Schlechtteil Vermeidung: Wurde ein Ausschussbehälter verwendet und ist die Ausschussklappe funktionfähig?	Wenn **Nein**, Los sperren	☐	☐	☐	☐	
Vorgabe: Entspricht die Kohlebürste der Zeichnung?	Wenn **Nein**, Los sperren	☐	☐	☐	☐	
Kabelunterstützung: Ist der Kabelrollenantrieb, die Kabelführung, die Kabelüberwachung und die Kabelbremse funktionfähig?	Wenn **Nein**, keine Produktionsfreigabe (Start) - Los Sperren (Ende)	☐	☐	☐	☐	☐
Kabelschere: Ist die Kabelschere (mit Federn & Scherenblättern) funktionfähig, die Führungsbuchse ohne Spiel und die Gewindestange in Ordnung?	Wenn **Nein**, keine Produktionsfreigabe (Start) - Los Sperren (Ende)	☐	☐	☐	☐	☐
Schweißr Ist die We t Keramiken) und ist die Sch ekt eingestellt?	Pro t) - Los Sperren (Ende)	☐	☐	☐	☐	☐
Prüfungv Ist die Kabellängenabfrage korrekt eingestellt und ist sie funktionfähig?	Wenn **Nein**, keine Produktionsfreigabe (Start) - Los Sperren (Ende)	☐	☐	☐	☐	☐

Are the brass bushings functional , i.e. free of play ?

If no, no start of production

Update P-FMEA

Fehlerart	Besondere Merkmale	Fehler-Auswirkung	Fehler-Ursache	Fehler-Vermeidung	Fehler-Entdeckung	A	B	E	AxBxE=RPZ	Maßnahme	Verant-wortl.	Termin	A	B	E	AxBxE=RZ
Litze nach hinten eingepresst		Beschädigung der Kohlebürste im Bereich der Nut	Verschleiß der Messingbuchse, dadurch kann die Schere an der Traverse nicht korrekt eingestellt werden.	Kontrolle der Messingbuchsen auf Verschleiß 1x pro Los	Beschädigung nach dem Schleifen sichtbar	4	5	4	80	Aufnahme in die Checkliste: 1x pro Los werden die Messingbuchsen auf Verschleiß kontrolliert (Reklamation Nidec Polen 250703)	DB	16.07.2025	2	5	3	30

Wire pressed in at the rear

Damage to the carbon brush in the groove area

Wear on the brass bushing, which prevents the shears from being adjusted correctly on the crossbar.

Check brass bushings for wear once per batch

Damage visible after grinding

Added to the checklist: The brass bushings are checked for wear once per batch (Complaint Nidec Poland 250703)